Table 49. I/O current injection susceptibility(1)

Symbol	Description		Functional susceptibility		Unit
			Negative injection	Positive injection	
I_{INJ}	Injected current on pin	Any IO	5	NA	mA

1. Evaluated by characterization – Not tested in production.

5.3.13 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in *Table 50* are derived from tests performed under the conditions summarized in *Table 24: General operating conditions*. All I/Os are designed as CMOS- and TTL-compliant.

For information on GPIO configuration, refer to the application note AN4899 *STM32 GPIO configuration for hardware settings and low-power consumption*, available on the ST website www.st.com.

Table 50. I/O static characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{IL}(1)	I/O input low level voltage	All	2 V < V_{DDIO1} < 3.6 V	-	-	0.3 x V_{DDIO1}	V
V_{IH}(1)	I/O input high level voltage	All	2 V < V_{DDIO1} < 3.6 V	0.7 x V_{DDIO1}	-	-	V
V_{hys}(2)	I/O input hysteresis	-		-	200	-	mV
I_{lkg}(3)	Input leakage current(3)	$0 < V_{IN} \leq V_{DDIO1}$		-	-70	-	nA
		$V_{DDIO1} \leq V_{IN} \leq V_{DDIO1}$ +1 V		-	600	-	
		V_{DDIO1} +1 V $\leq V_{IN}$		-	150	-	
R_{PU}	Weak pull-up equivalent resistor (4)	$V_{IN} = V_{SS}$		25	40	55	kΩ
R_{PD}	Weak pull-down equivalent resistor(4)	V_{IN} = VDDIO1		25	40	55	kΩ
C_{IO}	I/O pin capacitance	-		-	5	-	pF

1. Refer to *Figure 19: I/O input characteristics*.
2. Specified by design – Not tested in production.
3. This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula: ITotal_Ileak_max = 10 µA + [number of I/Os where VIN is applied on the pad] x Ilkg(Max).
4. Pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).